Test sets, statistics and results

COS30049 A2, group HCMC_COS30049.1_4 (generated by `tools/test_report.py`)

2026-10-26 22:25 UTC, commit `46efdd3`

What is measured

Every writer answers each prompt; code then checks the answer (`explorer/assistant.py`, `lm/eval_explain.py`). *Guardrail*: numbers only from the facts, comparison words that match the data, the right verdict (flagged, not flagged, not scored), the limitation line on every flag, no disallowed wording. *Fully correct*: the guardrail passes and the answer names the template's reasons, strongest first, each number on its own feature's line (features whose $|z|$ shows the same in the prompt may come in either order). *Macro*: the mean over cases, so common easy cases cannot carry the rate. *Nothing out*: fully correct where no feature stands out, the one case in which a writer must rank features itself. All figures are percentages. The template is the reference, so it scores 100 by construction; it checks that each set is consistent.

Overview: fully correct, by test set

Test set	Prompts	Template	152M r1	Qwen r1	Qwen r2	152M r2
DS7 v1 test (frozen, round 1)	445	100.0	87.0	95.5	96.2	–
DS10 (teammate's snapshot)	291	100.0	83.2	95.9	96.6	–
Random: real	500	100.0	88.4	95.4	95.0	–
Random: synthetic	500	100.0	77.2	94.2	97.6	–
Random: hard	500	100.0	43.2	77.4	82.0	–
Random: balanced	883	100.0	72.0	86.0	97.1	–
Crawled: DS1 test split	174	100.0	89.7	93.7	95.4	–
Crawled: DS2 test split	421	100.0	82.4	97.1	97.9	–
Crawled: DS10	334	100.0	76.3	97.3	98.2	–
Close calls	300	100.0	80.0	92.3	93.7	–
Extreme values	300	100.0	74.7	95.3	96.7	–
Maximal shift	300	100.0	18.7	46.7	66.3	–

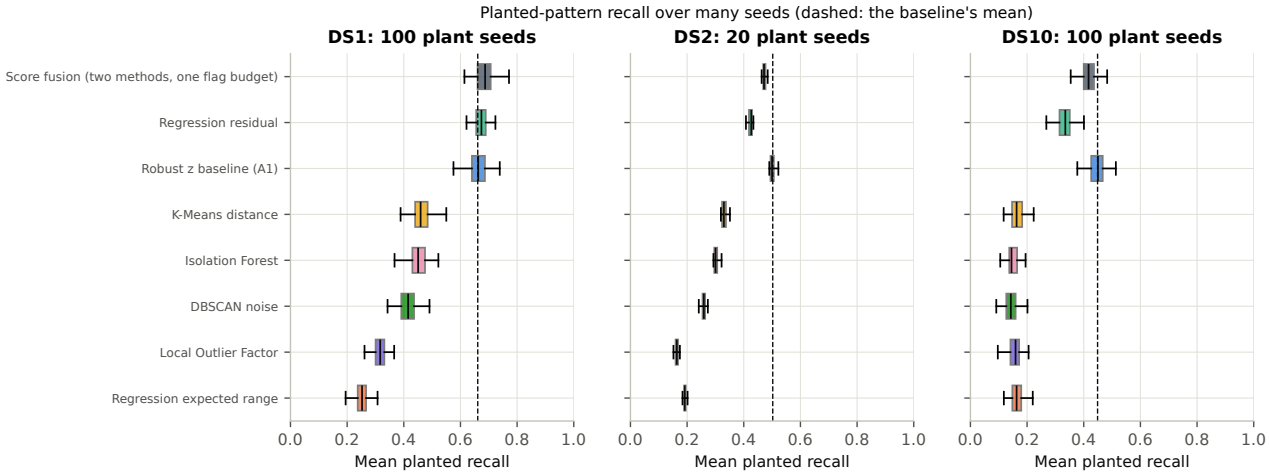
Statistics campaign

Distributions rather than single runs (requested 27 Oct). The scoring methods' planted patterns were replanted with many seeds (`explorer/distribution.py`); the classifiers were refitted with ten seeds and their test scores bootstrapped 1,000 times (`explorer/classify_distribution.py`). The language models were scored on 12 test sets with intervals and paired tests (extttlm/eval_stats.py).

Scoring methods: mean planted recall over plant seeds

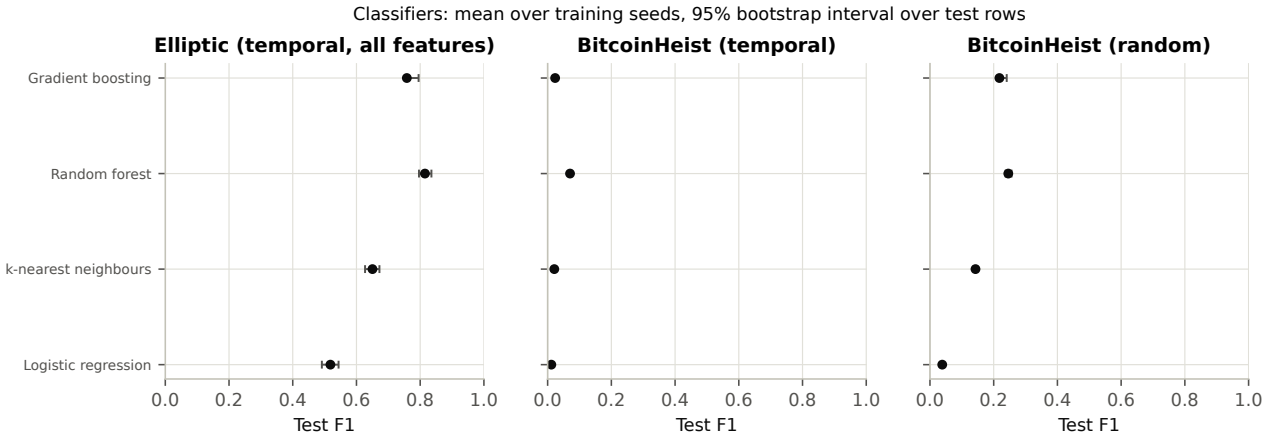
Data	Method	Seeds	Mean (sd)	95% of seeds	Beats baseline	Mean rank
DS1	Score fusion	100	0.685 (0.034)	0.621–0.745	95%	1.4
DS1	Regression residual	100	0.673 (0.024)	0.630–0.716	59%	2.0
DS1	Robust-z baseline	100	0.661 (0.035)	0.588–0.720	0%	2.5
DS1	K-Means distance	100	0.463 (0.034)	0.403–0.534	0%	4.4
DS1	Isolation Forest	100	0.452 (0.036)	0.386–0.522	0%	4.8
DS1	DBSCAN noise	100	0.415 (0.034)	0.357–0.483	0%	5.8

Data	Method	Seeds	Mean (sd)	95% of seeds	Beats baseline	Mean rank
DS1	Local Outlier Factor	100	0.316 (0.023)	0.271–0.357	0%	7.0
DS1	Regression range	100	0.252 (0.026)	0.204–0.301	0%	8.0
DS10	Robust-z baseline	100	0.449 (0.031)	0.381–0.507	0%	1.0
DS10	Score fusion	100	0.416 (0.032)	0.349–0.476	1%	2.0
DS10	Regression residual	100	0.333 (0.028)	0.278–0.389	0%	3.0
DS10	K-Means distance	100	0.165 (0.026)	0.122–0.217	0%	5.3
DS10	Regression range	100	0.165 (0.024)	0.121–0.208	0%	5.5
DS10	Local Outlier Factor	100	0.158 (0.024)	0.117–0.204	0%	5.8
DS10	Isolation Forest	100	0.149 (0.023)	0.108–0.193	0%	6.5
DS10	DBSCAN noise	100	0.145 (0.027)	0.104–0.207	0%	6.9
DS2	Robust-z baseline	20	0.502 (0.010)	0.491–0.524	0%	1.0
DS2	Score fusion	20	0.475 (0.010)	0.464–0.496	0%	2.0
DS2	Regression residual	20	0.424 (0.007)	0.410–0.433	0%	3.0
DS2	K-Means distance	20	0.332 (0.009)	0.320–0.350	0%	4.0
DS2	Isolation Forest	20	0.303 (0.009)	0.293–0.319	0%	5.0
DS2	DBSCAN noise	20	0.259 (0.009)	0.244–0.277	0%	6.0
DS2	Regression range	20	0.192 (0.006)	0.181–0.200	0%	7.0
DS2	Local Outlier Factor	20	0.164 (0.007)	0.152–0.174	0%	8.0



Classifiers: test F1 over training seeds, with a bootstrap of the test rows

Data and split	Model	Seeds	F1 mean (seed sd)	Bootstrap 95% CI, first seed	PR-A
Elliptic (temporal, all features)	Logistic regression	1	0.518 (0.000)	0.491–0.544	0
Elliptic (temporal, all features)	k-nearest neighbours	1	0.650 (0.000)	0.627–0.672	0
Elliptic (temporal, all features)	Random forest	10	0.815 (0.002)	0.796–0.835	0
Elliptic (temporal, all features)	Gradient boosting	10	0.758 (0.013)	0.755–0.795	0
BitcoinHeist (temporal)	Logistic regression	1	0.012 (0.000)	0.011–0.012	0
BitcoinHeist (temporal)	k-nearest neighbours	1	0.021 (0.000)	0.020–0.022	0
BitcoinHeist (temporal)	Random forest	10	0.070 (0.000)	0.068–0.075	0
BitcoinHeist (temporal)	Gradient boosting	10	0.024 (0.000)	0.023–0.024	0
BitcoinHeist (random)	Logistic regression	1	0.039 (0.000)	0.037–0.039	0
BitcoinHeist (random)	k-nearest neighbours	1	0.143 (0.000)	0.137–0.148	0
BitcoinHeist (random)	Random forest	10	0.246 (0.001)	0.238–0.256	0
BitcoinHeist (random)	Gradient boosting	10	0.218 (0.008)	0.224–0.241	0

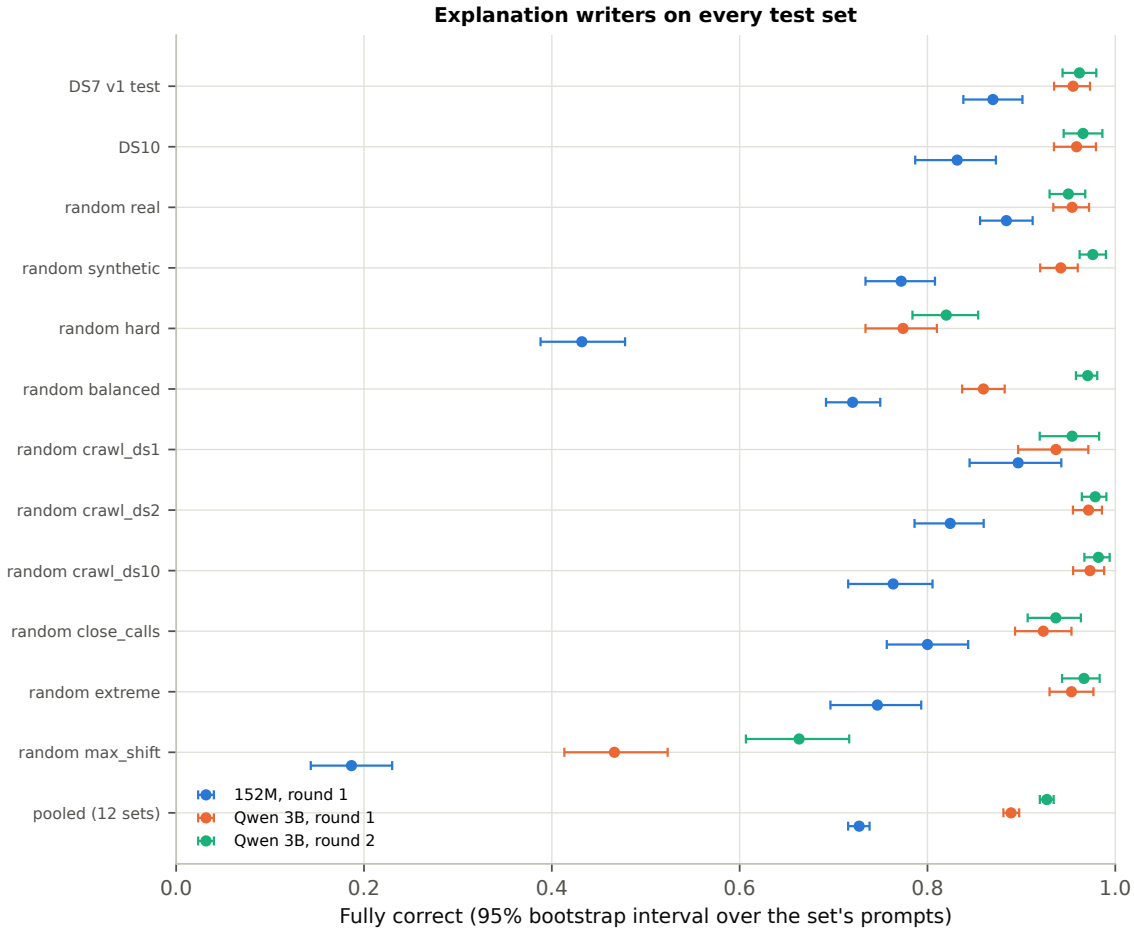


Explanation writers: fully correct with 95% bootstrap intervals, and paired tests

Test set	Writer	Rate	95% CI
pooled (12 sets)	152M, round 1	72.7	71.6–73.8
pooled (12 sets)	Qwen 3B, round 1	88.9	88.1–89.8
pooled (12 sets)	Qwen 3B, round 2	92.7	92.0–93.4
glossary (key facts)	Qwen 3B, round 1	15.3	9.5–22.1
glossary (key facts)	Qwen 3B, round 2	41.3	31.3–50.9

Qwen round 1 minus round 2 on identical prompts (negative: round 2 better); McNemar's exact test counts the prompts only one of them got right.

Test set	Difference	95% CI	Only r1 right	Only r2 right	McNemar p
DS7 v1 test	-0.7	-2.5 to +0.9	6	9	0.6072
DS10	-0.7	-2.7 to +1.4	4	6	0.7539
random real	+0.4	-1.0 to +1.8	7	5	0.7744
random synthetic	-3.4	-5.2 to -1.8	3	20	0.0005
random hard	-4.6	-8.6 to -0.8	38	61	0.0265
random balanced	-11.1	-13.1 to -8.9	6	104	<0.0001
random crawl_ds1	-1.7	-4.6 to +0.6	1	4	0.3750
random crawl_ds2	-0.7	-2.6 to +1.0	5	8	0.5811
random crawl_ds10	-0.9	-2.7 to +0.9	3	6	0.5078
random close_calls	-1.3	-3.7 to +1.3	5	9	0.4240
random extreme	-1.3	-3.3 to +0.3	2	6	0.2891
random max_shift	-19.7	-25.7 to -13.0	28	87	<0.0001
glossary (key facts)	-25.9	-35.8 to -16.2	–	–	–



The language models' test sets

DS7 v1 test (frozen, round 1)

Source. DS1 and DS2 transactions of the test-split addresses (DS7 splits by a seeded hash of the address).
How drawn. Round 1's DS7 build: up to 40 pairs per DS1 address and 10 per DS2 address, half flagged at most.
File. `data/processed/ds7/test.jsonl` (445 prompts)

Case	Prompts
unscored	39
flagged, 1 reason	64
flagged, 2 reasons	58
flagged, 3 reasons	28
flagged, 4 reasons	6
not flagged, nothing stands out	167
not flagged, 1 reason	62
not flagged, 2 reasons	16
not flagged, 3 reasons	5

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.3	100.0	97.8	87.6	87.0	72.7	86.8
Qwen 3B, round 1	98.9	100.0	99.8	95.7	95.5	96.9	88.6
Qwen 3B, round 2	99.8	100.0	100.0	96.4	96.2	98.9	89.8

Not yet scored on this set: 152M, round 2.

DS10 (teammate's snapshot)

Source. DS10: Le Quang Anh Hoang's independent Esplora snapshot (27 addresses), unseen by every model.
How drawn. The DS7 rule applied to DS10 by 'explorer.cli ds10-check'; frozen.
File. `data/processed/ds10/explain.jsonl` (291 prompts)

Case	Prompts
unscored	27
flagged, 1 reason	26
flagged, 2 reasons	54
flagged, 3 reasons	27
flagged, 4 reasons	2
not flagged, nothing stands out	102
not flagged, 1 reason	36
not flagged, 2 reasons	13
not flagged, 3 reasons	3
not flagged, 4 reasons	1

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.3	100.0	97.6	83.8	83.2	72.5	81.4
Qwen 3B, round 1	98.6	100.0	98.6	96.2	95.9	88.9	89.2
Qwen 3B, round 2	99.3	100.0	100.0	97.3	96.6	94.1	91.2

Not yet scored on this set: 152M, round 2.

Random: real

Source. DS7 v2 test split: DS1 and DS2 test-split addresses.
How drawn. 500 drawn at random (seed 50000), none in the frozen v1 test.
File. `data/processed/random_test/real.jsonl` (500 prompts)

Case	Prompts
unscored	33
flagged, 1 reason	47
flagged, 2 reasons	50
flagged, 3 reasons	23
flagged, 4 reasons	2
not flagged, nothing stands out	256
not flagged, 1 reason	66
not flagged, 2 reasons	20
not flagged, 3 reasons	3

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.2	100.0	97.6	89.2	88.4	85.4	84.0
Qwen 3B, round 1	99.4	100.0	99.8	95.8	95.4	99.0	91.0
Qwen 3B, round 2	99.6	100.0	100.0	95.2	95.0	98.9	90.2

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-random-test.log`, last run):

```
2026-10-26 21:32:39 INFO Step build-random-test started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-random-test
2026-10-26 21:32:40 INFO Randomized test set: 500 real prompts (from 1401 eligible), 500 synthetic and 500 hard: {'real': {'flagged': 122, 'not_fl
2026-10-26 21:32:40 INFO Step build-random-test finished in 1 s (exit 0)
```

Random: synthetic

Source. Made up: values, usual values and z drawn from settings randomTest, capped to realistic limits.
How drawn. 500 transactions (seed 50000); z from a mixture, values derived from z.
File. `data/processed/random_test/synthetic.jsonl` (500 prompts)

Case	Prompts
unscored	34
flagged, 1 reason	36
flagged, 2 reasons	70
flagged, 3 reasons	54
flagged, 4 reasons	27
not flagged, nothing stands out	79
not flagged, 1 reason	116
not flagged, 2 reasons	62
not flagged, 3 reasons	16
not flagged, 4 reasons	6

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	96.8	100.0	96.6	78.2	77.2	66.2	77.2
Qwen 3B, round 1	99.0	100.0	96.6	95.0	94.2	82.7	96.2
Qwen 3B, round 2	99.2	100.0	99.2	98.2	97.6	93.7	97.5

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-random-test.log`, last run):

```
2026-10-26 21:32:39 INFO Step build-random-test started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-random-test
2026-10-26 21:32:40 INFO Randomized test set: 500 real prompts (from 1401 eligible), 500 synthetic and 500 hard: {'real': {'flagged': 122, 'not_fl
2026-10-26 21:32:40 INFO Step build-random-test finished in 1 s (exit 0)
```

Random: hard

Source. Made up, explained for any of the nine methods.
How drawn. 500 (seed 50001): 289 non-baseline, about half reworded and half shuffled.
File. `data/processed/random_test/hard.jsonl` (500 prompts)

Case	Prompts
unscored	44
flagged, nothing stands out	13
flagged, 1 reason	46
flagged, 2 reasons	60
flagged, 3 reasons	38
flagged, 4 reasons	14
not flagged, nothing stands out	73
not flagged, 1 reason	105
not flagged, 2 reasons	69
not flagged, 3 reasons	29
not flagged, 4 reasons	9

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	85.2	99.6	57.2	46.0	43.2	33.7	44.2
Qwen 3B, round 1	92.6	97.2	89.2	83.2	77.4	61.1	62.8
Qwen 3B, round 2	99.0	99.8	85.0	82.6	82.0	69.4	74.4

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-random-test.log`, last run):

```
2026-10-26 21:32:39 INFO Step build-random-test started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-random-test
2026-10-26 21:32:40 INFO Randomized test set: 500 real prompts (from 1401 eligible), 500 synthetic and 500 hard: {'real': {'flagged': 122, 'not_fl
2026-10-26 21:32:40 INFO Step build-random-test finished in 1 s (exit 0)
```

Random: balanced

Source. Every transaction of DS10 and of DS2's 173 test-split addresses (46,513 explained).
How drawn. De-duplicated, other test sets removed, up to 100 per case (seed 50002).
File. `data/processed/random_test/balanced.jsonl` (883 prompts)

Case	Prompts
unscored	1
flagged, 1 reason	100
flagged, 2 reasons	100
flagged, 3 reasons	100
flagged, 4 reasons	95
not flagged, nothing stands out	100
not flagged, 1 reason	100
not flagged, 2 reasons	100
not flagged, 3 reasons	100
not flagged, 4 reasons	87

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.3	100.0	93.1	72.4	72.0	73.9	82.0
Qwen 3B, round 1	98.8	100.0	88.1	86.4	86.0	86.2	96.0
Qwen 3B, round 2	99.7	100.0	99.5	97.3	97.1	97.2	93.0

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-balanced-test.log`, last run):

```
2026-10-26 21:43:37 INFO Step build-balanced-test started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-balanced-test
2026-10-26 21:43:48 INFO DS2 test split: 173 addresses, 40728 transactions explained
2026-10-26 21:43:50 INFO DS10: 27 addresses, 5785 transactions explained
2026-10-26 21:43:51 INFO Balanced test set: 883 prompts; per case (unique in pool, drawn): {'flagged, 1 reason': (1426, 100), 'flagged, 2 reasons'
2026-10-26 21:43:51 INFO Step build-balanced-test finished in 13 s (exit 0)
```

Crawled: DS1 test split

Source. DS1 (our Esplora snapshot): the test-split addresses' transactions.
How drawn. Up to 60 per case (seed 50003); no prompt of an earlier set.
File. `data/processed/random_test/crawl_ds1.jsonl` (174 prompts)

Case	Prompts
unscored	1
flagged, 1 reason	11
flagged, 2 reasons	7
flagged, 3 reasons	4
flagged, 4 reasons	2
not flagged, nothing stands out	60
not flagged, 1 reason	60
not flagged, 2 reasons	26
not flagged, 3 reasons	3

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.4	100.0	98.9	90.2	89.7	80.6	86.7
Qwen 3B, round 1	99.4	100.0	99.4	94.3	93.7	98.0	81.7
Qwen 3B, round 2	99.4	100.0	100.0	96.0	95.4	98.5	86.7

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Crawled: DS2 test split

Source. DS2 (our Esplora crawl of 4,012 addresses): the 173 test-split addresses.
How drawn. Up to 60 per case (seed 50003); no prompt of an earlier set.
File. `data/processed/random_test/crawl_ds2.jsonl` (421 prompts)

Case	Prompts
unscored	1
flagged, 1 reason	60
flagged, 2 reasons	60
flagged, 3 reasons	60
not flagged, nothing stands out	60
not flagged, 1 reason	60
not flagged, 2 reasons	60
not flagged, 3 reasons	60

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	99.8	100.0	99.3	82.9	82.4	84.6	86.7
Qwen 3B, round 1	98.3	100.0	99.8	98.3	97.1	97.5	86.7
Qwen 3B, round 2	99.5	100.0	100.0	98.3	97.9	98.1	93.3

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Crawled: DS10

Source. DS10 (the teammate's crawl): all 27 addresses.
How drawn. Up to 60 per case (seed 50003); no prompt of an earlier set.
File. `data/processed/random_test/crawl_ds10.jsonl` (334 prompts)

Case	Prompts
unscored	1
flagged, 1 reason	30
flagged, 2 reasons	48
flagged, 3 reasons	15
not flagged, nothing stands out	60
not flagged, 1 reason	60
not flagged, 2 reasons	60
not flagged, 3 reasons	60

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	100.0	100.0	97.3	76.3	76.3	80.0	78.3
Qwen 3B, round 1	99.4	100.0	99.4	97.6	97.3	98.1	86.7
Qwen 3B, round 2	100.0	100.0	100.0	98.2	98.2	98.8	90.0

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
```

```
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Close calls

Source. DS1, DS2 and DS10 test-split transactions.
How drawn. 300 whose listed |z| differ by less than 0.3, so order and pair are hardest.
File. `data/processed/random_test/close_calls.jsonl` (300 prompts)

Case	Prompts
flagged, 3 reasons	2
not flagged, nothing stands out	276
not flagged, 2 reasons	17
not flagged, 3 reasons	5

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	97.7	100.0	94.3	81.7	80.0	54.2	82.2
Qwen 3B, round 1	98.7	100.0	98.7	93.0	92.3	93.0	92.0
Qwen 3B, round 2	99.3	100.0	100.0	94.0	93.7	98.3	93.1

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons':
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Extreme values

Source. Made up, one extreme regime per transaction.
How drawn. 300: 1-100 sat or about 1,000 BTC amounts, minute or multi-year gaps, 80-200 transactions a week.
File. `data/processed/random_test/extreme.jsonl` (300 prompts)

Case	Prompts
unscored	23
flagged, 1 reason	16
flagged, 2 reasons	44
flagged, 3 reasons	33
flagged, 4 reasons	5
not flagged, nothing stands out	64
not flagged, 1 reason	61
not flagged, 2 reasons	45
not flagged, 3 reasons	7
not flagged, 4 reasons	2

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	89.0	100.0	99.0	83.3	74.7	68.1	70.3
Qwen 3B, round 1	98.7	100.0	98.3	96.0	95.3	81.2	90.6
Qwen 3B, round 2	98.3	100.0	100.0	98.0	96.7	91.7	90.6

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
```

```
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons'
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons'
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Maximal shift

Source. Made up, for methods other than the baseline.
How drawn. 300: every prompt reworded, its facts shuffled, the flag independent of the features.
File. `data/processed/random_test/max_shift.jsonl` (300 prompts)

Case	Prompts
unscored	28
flagged, nothing stands out	18
flagged, 1 reason	24
flagged, 2 reasons	21
flagged, 3 reasons	17
flagged, 4 reasons	4
not flagged, nothing stands out	40
not flagged, 1 reason	50
not flagged, 2 reasons	51
not flagged, 3 reasons	37
not flagged, 4 reasons	10

Writer	Guardrail	Verdict	Reasons	Order	Fully	Macro	Nothing out
Template (control)	100.0	100.0	100.0	100.0	100.0	100.0	100.0
152M, round 1	72.3	96.7	27.3	20.0	18.7	14.6	8.6
Qwen 3B, round 1	73.3	86.7	72.0	64.7	46.7	36.1	25.9
Qwen 3B, round 2	97.3	100.0	73.3	68.7	66.3	62.7	48.3

Not yet scored on this set: 152M, round 2.

Build log (`reports/logs/build-more-tests.log`, last run):

```
2026-10-27 00:09:07 INFO Step build-more-tests started: C:\Users\ADMIN\COS30049A2\explorer\cli.py build-more-tests
2026-10-27 00:09:21 INFO DS1 test split: 522 transactions explained
2026-10-27 00:09:21 INFO DS2 test split: 40728 transactions explained
2026-10-27 00:09:21 INFO DS10: 5785 transactions explained
2026-10-27 00:09:22 INFO crawl_ds1: 174 prompts; {'flagged, 1 reason': 11, 'flagged, 2 reasons': 7, 'flagged, 3 reasons': 4, 'flagged, 4 reasons'
2026-10-27 00:09:22 INFO crawl_ds2: 421 prompts; {'flagged, 1 reason': 60, 'flagged, 2 reasons': 60, 'flagged, 3 reasons': 60, 'not flagged, 1 rea
2026-10-27 00:09:22 INFO crawl_ds10: 334 prompts; {'flagged, 1 reason': 30, 'flagged, 2 reasons': 48, 'flagged, 3 reasons': 15, 'not flagged, 1 re
2026-10-27 00:09:22 INFO close_calls: 300 prompts; {'flagged, 3 reasons': 2, 'not flagged, 2 reasons': 17, 'not flagged, 3 reasons': 5, 'not flagg
2026-10-27 00:09:22 INFO extreme: 300 prompts; {'flagged, 1 reason': 16, 'flagged, 2 reasons': 44, 'flagged, 3 reasons': 33, 'flagged, 4 reasons'
2026-10-27 00:09:22 INFO max_shift: 300 prompts; {'flagged, 1 reason': 24, 'flagged, 2 reasons': 21, 'flagged, 3 reasons': 17, 'flagged, 4 reasons
2026-10-27 00:09:22 INFO Step build-more-tests finished in 15 s (exit 0)
```

Glossary test (open questions)

Source. `tools/make_glossary.py`: 74 held-out phrasings of the 65 taught topics, each with the key facts a correct answer must state.
How drawn. One or two per topic, written differently from the taught questions; round 1's five test questions are among them.

Writer	Key facts stated	Fully right	ROUGE-L
Qwen 3B, round 1	15.3	6.8	0.200
Qwen 3B, round 2	41.3	31.1	0.449